To what extent do GPT language models capture human sentence processing dynamics? Evidence from Korean subject honorification

Gyu-Ho Shin (University of Illinois Chicago; ghshin@uic.edu), Nayoung Kwon (University of Oregon), Seongmin Mun (Kyungpook National University)

Understanding whether large language models (LLMs) capture human sentence-processing dynamics remains central to current debates in computational psycholinguistics. Korean subject honorification offers a particularly stringent test case: it encodes morphosyntactic agreement while simultaneously indexing socio-pragmatic stance, thereby requiring the integration of grammatical and semantic-pragmatic knowledge. Prior work has shown that LLMs often track categorical violations but struggle with gradience and pragmatic nuance in English. However, empirical evidence from typologically under-explored languages remains scarce, limiting generalisability of English-based findings. The present study therefore examines the extent to which three Korean GPT variants approximate human sentence-processing patterns in two domains—politeness rating and self-paced reading—drawing upon open-access datasets. Our overarching goal is to assess how faithfully surprisal derived from autoregressive language models aligns with human sensitivity to honorific agreement, animacy, and the pragmatic felicity of honorific marking.

We analysed two complementary datasets from Korean natives (Table1): (i) a politeness-rating task (1) where participants evaluated sentences varying in subject honorifiability (e.g., teacher vs. kid), predicate honorific marking, and subject animacy (e.g., book), and (ii) a self-paced reading task (2) where the critical subject region followed an honorific or non-honorific predicate across a clausal boundary. For each dataset, we compared human ratings or reading times with word- and region-level surprisal estimated from three Korean GPT models (Table2). Statistical analyses employed ordinal regression with cumulative-link mixed models for ratings and linear mixed-effects models for reading times, with parallel models fitted to GPT surprisal. Correlations between human measures and model surprisal were assessed on an item-wise basis.

Results: Politeness rating (Fig1). Human ratings displayed a clear pragmatic hierarchy: sentences combining honorific predicates with honorifiable human subjects were judged most polite, while those involving inanimate subjects were judged least polite, even when syntactically well-formed. This animacy-based scaling was robust and systematic. By contrast, GPT lacked sensitivity to animacy. Across models, surprisal was driven primarily by overt morphosyntactic mismatch (e.g., non-honorifiable subject + honorific predicate), with inanimate subjects often treated similarly to animate non-honorifiable subject. Item-wise correlations between human politeness ratings and surprisal did not reach significance. These findings indicate that GPT models lack the socio-pragmatic sensitivity that human participants deploy when interpreting honorific marking.

Results: Self-paced reading (Fig2). Human readers exhibited maximal processing difficulty at the critical subject region when a preceding honorific predicate was followed by a non-honorifiable, especially inanimate, subject—demonstrating sensitivity to both morphosyntactic and animacy-based constraints. GPT surprisal captured coarse patterns of mismatch but did not reproduce this animacy modulation: all models treated animate and inanimate mismatches similarly and often attributed additional surprisal to subject types that elicited little human slowdown. No significant correlations emerged between human reading times and model surprisal.

Taken together, these findings show that GPT models can capture coarse grammatical anomalies in Korean honorification but fail to replicate the gradient, context-dependent processing patterns observed in humans. The models' insensitivity to animacy and pragmatic felicity highlights the limits of next-token prediction as a mechanistic model of human sentence processing, indicating that social and pragmatic knowledge is not recoverable from token-wise distributional statistics alone. Our results therefore suggest that advancing computational models of sentence processing will require enriched frameworks that integrate cultural, semantic, and discourse-level information alongside surface predictability.

Table 1. Subject−predicate combinations used for politeness rating and self-paced reading tasks

Subject	Predicate	Acceptable?	*Note*. Two control conditions with an inanimate subject were included for each predicate type, creating semantic anomaly conditions.
H_{subj}	H_{pred}	Yes	
	NH_{pred}	Yes	
NH_{subj}	H_{pred}	No (Structural anomaly)	
	NH_{pred}	Yes	
INA_{subj}	H_{pred}	No (Semantic anomaly)	
	NH_{pred}	No (Semantic anomaly)	

(1) sensayngnim-kkeyse/kkoma-ka/chayk-i pang-ulo tuleka-(si)-ess-ta.
teacher-NOM.HON/kid-NOM/book-NOM room-DIR enter-(SH)-PST-DCL
'The teacher/kid/book entered the room.'

(2) pang-ulo [tuleka-(si)-mye]$_{R3}$ [sensayngnim-kkeyse/kkoma-ka/chayk-i]$_{R4}$ …
room-DIR [enter-(SH)-while]$_{R3}$ [teacher-NOM.HON/kid-NOM/book-NOM]$_{R4}$ …
'While the teacher/kid/book entered the room, …'

Abbreviation. DCL: predicate ending (declarative); DIR: directional marker; HON: honorific feature; NOM: nominative case marker; PST: past tense marker; SH: subject honorific marker; SUBJ: subject.

Table 2. Information about three GPT models

	Number of parameters	Number of layers	Number of heads	Hidden layer dim	Feedforward network dim
KoGPT-2	125M	12	12	768	3,072
Ko-GPT-Trinity	1.2B	24	16	2,048	8,192
mGPT	1.3B	24	16	2,048	8,192

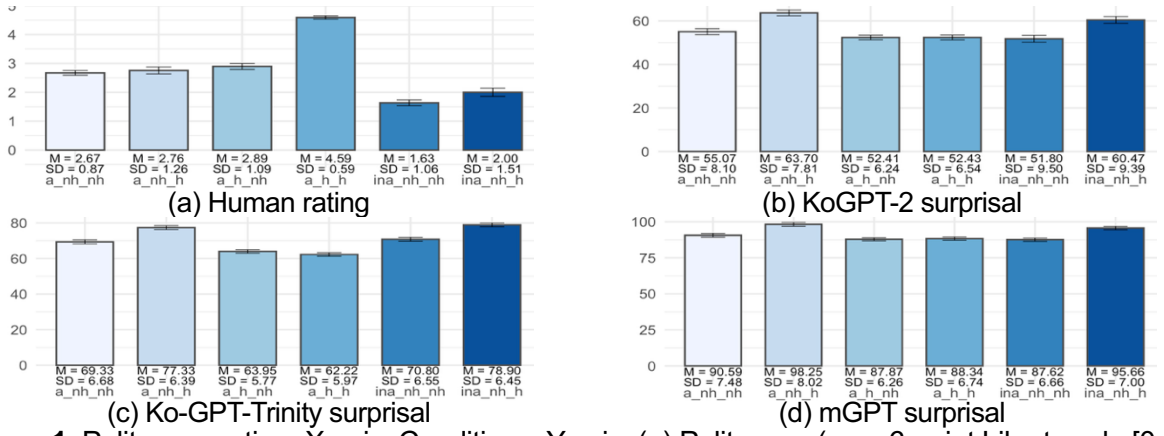

Figure 1. Politeness rating. X-axis: Conditions; Y-axis: (a) Politeness (on a 6-point Likert scale [0 = "very impolite," 5 = "very polite"]), (b–d) Surprisal. Error bars indicate 95% CIs.

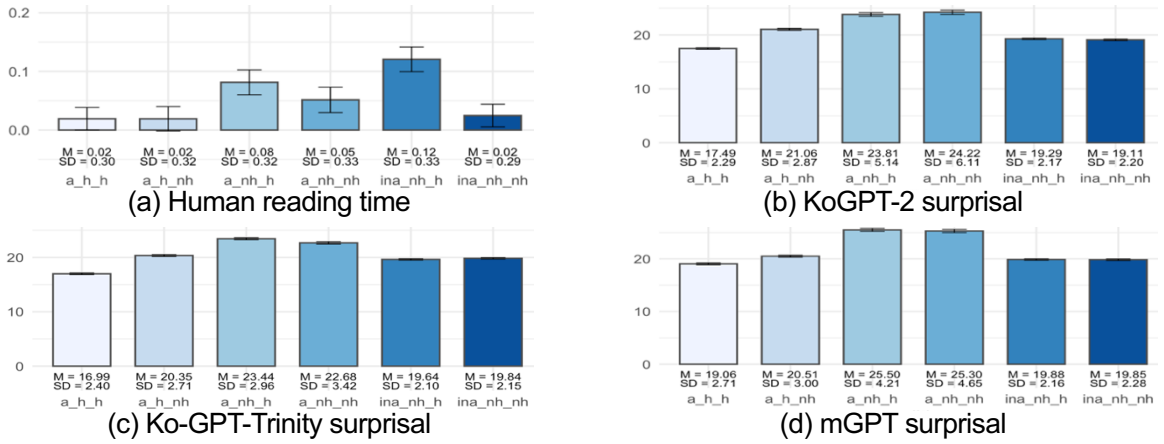

Figure 2. Self-paced reading (R4). X-axis: Condition; Y-axis: (a) Residual reading time, (b–d) Surprisal. Error bars indicate 95% CIs.